Part1e:

a. Based on the results, the distilled student model has the smallest size. The pruned model has the best performance by having a accuracy of 0.97
b. I have combined all the techniques together to shrink the model size(knowledge distillation first and then prune the student model. Undergoes a int8 quantization at last)

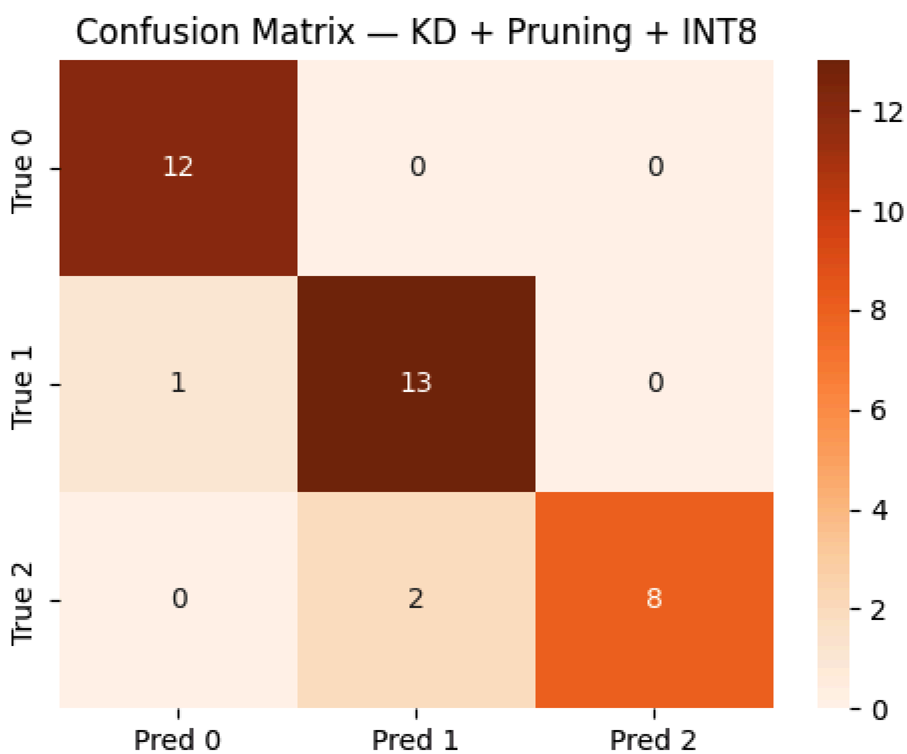
```
Final model saved → 'model_final.tflite'
File size: 3.71 KB

── Final Model (KD + Pruning + INT8) ──────────────────────
              precision    recall  f1-score   support

     Class 0       0.92      1.00      0.96        12
     Class 1       0.87      0.93      0.90        14
     Class 2       1.00      0.80      0.89        10

    accuracy                           0.92        36
   macro avg       0.93      0.91      0.92        36
weighted avg       0.92      0.92      0.92        36
```

c.



Based on the result, the final model has the smallest size(3.71KB) while only have a small decrease in terms of accuracy(0.97 to 0.92)

Part2

a.

i. Supervised — Classification:

1. Classification (Keras) — custom NN classifier (e.g., motion/gesture recognition from IMU data)
2. Regression (Keras) — predicts a continuous value (e.g., estimating temperature from sensor data)
3. Image Classification via Transfer Learning — fine-tunes MobileNet, etc. (e.g., visual defect detection)
4. Keyword Spotting via Transfer Learning — fine-tunes audio models (e.g., wake-word detection)
5. Object Detection (MobileNetV2 SSD FPN / FOMO) — locates objects in images (e.g., person detection)

ii. Unsupervised — Anomaly Detection:

1. Anomaly Detection (K-means) — clusters normal behavior; flags outliers (e.g., machine vibration anomaly)
2. Anomaly Detection (GMM) — uses Gaussian Mixture Models for anomaly scoring
3. Visual Anomaly Detection (FOMO-AD) — detects visual anomalies without labeled defect data

iii. Classical ML: Classical ML block — supports scikit-learn algorithms for classification and regression (e.g., Random Forest, SVM, k-NN).

b.

i. Dense: Connects every input neuron to every output neuron; the core layer for learning feature-to-class mappings

ii. Reshape: Changes the shape of input data without altering its contents; used to prepare data for layers requiring a specific input shape

iii. Flatten: Converts multi-dimensional data into a 1D array

iv. Dropout: A regularization technique that randomly sets a fraction of inputs to zero during training, reducing overfitting

v. 1D convolution: Designed for sequential/time-series data (e.g., audio, accelerometer); applies learned filters along the time axis to extract local features

vi. 2D convolution: Applies 2D filters over spatial data (e.g., spectrograms, images) to detect local patterns such as edges or textures

vii. Maxpooling: Downsamples feature maps by taking the maximum value in a window, reducing spatial size and computation

viii. BatchNormalization: Normalizes activations within a mini-batch to stabilize and accelerate training

c.

i. Model Precision (Quantization):

1. Quantized (int8) — reduces weight precision from float32 to 8-bit integers, cutting memory and inference cost with minimal accuracy loss
2. Unoptimized (float32) — full-precision baseline model

ii. Compiler Options:

1. TFLite (TensorFlow Lite for Microcontrollers) — standard interpreter-based runtime
2. EON Compiler — interpreter-less code generator that compiles the neural network directly to C++ kernels, reducing RAM and flash usage while maintaining accuracy
3. EON Compiler (RAM optimized) — an additional variant that reduces RAM even further when architecturally possible

d.

i. Edge Impulse supports two input modalities for synthetic data generation: images and audio. For images, users can use DALL-E (GPT-4) to generate realistic images from text prompts. Black Forest Labs FLUX-Pro is also available as a second image generation option, offering hyper-realistic output with strong prompt following and high output diversity, hosted via Replicate.

ii. For audio, OpenAI Whisper generates human-like speech samples suited for keyword spotting datasets, while ElevenLabs Sound Effects produces audible events such as glass breaking or alarm sounds for audio event detection tasks. Enterprise users can additionally connect custom LLM sources or self-hosted models via transformation blocks.

iii. Synthetic data is important in ML because it mimics real-world examples without requiring manual collection, making it especially valuable when real data is scarce, hard to obtain, or privacy-sensitive.